

May internship goals

01 Guest program memory usage

some of the [runs on EEST are crashing](#) due to the memory limit of zkVM

- *memory is purposefully small*
- *default ZisK allocator never frees*

*Result: resolved 13 tests for Ethrex
(up to 60M gas)*

02 More optimizations

unfinished ideas from the previous month and optimization for the LLVM arguments

Result: combined speed up ~13%

Current changes

[Ethrex #6664: Merged](#)

*fixes ~11 previously
crashing EEST tests*

fix(l1): resolve memory limit crash on zisk
reuse already allocated callframe memory

[Ethrex #6708: In review](#)

*fixes 2 previously
crashing EEST tests*

perf(l1): replace index based jump targets with bit vector
same approach as Reth, Geth
uses ~10% less memory on average for mainnet blocks

[Ethrex #6774: In review](#)

measured speed-up: ~1.5%

perf(l1): route remaining hash calls through precompile#6774
refactor to use precompiles for (*almost*) all hash function calls

Current changes

[Ethrex #6797: In review](#)

feat(l1): serialize execution witness with ssz instead of rkyv

move the serialization closer to [the specification](#)

state is stored as flat SSZ nodes and tree needs to be reconstructed in the guest

[ere #374: Merged](#)

perf: add optimized llvm parameters for the zisk target

based on the paper [Evaluating Compiler Optimization Impacts on zkVM Performance](#)

also check my [write-up with findings](#)

*speed-up: ~10.5%
requires no code change*

Next steps

01 Compare allocators on ZisK

ZisK currently offers: bump, dlmalloc, talc, tifs

02 More profiling on Ethrex

identify remaining zkVM-unfriendly patterns